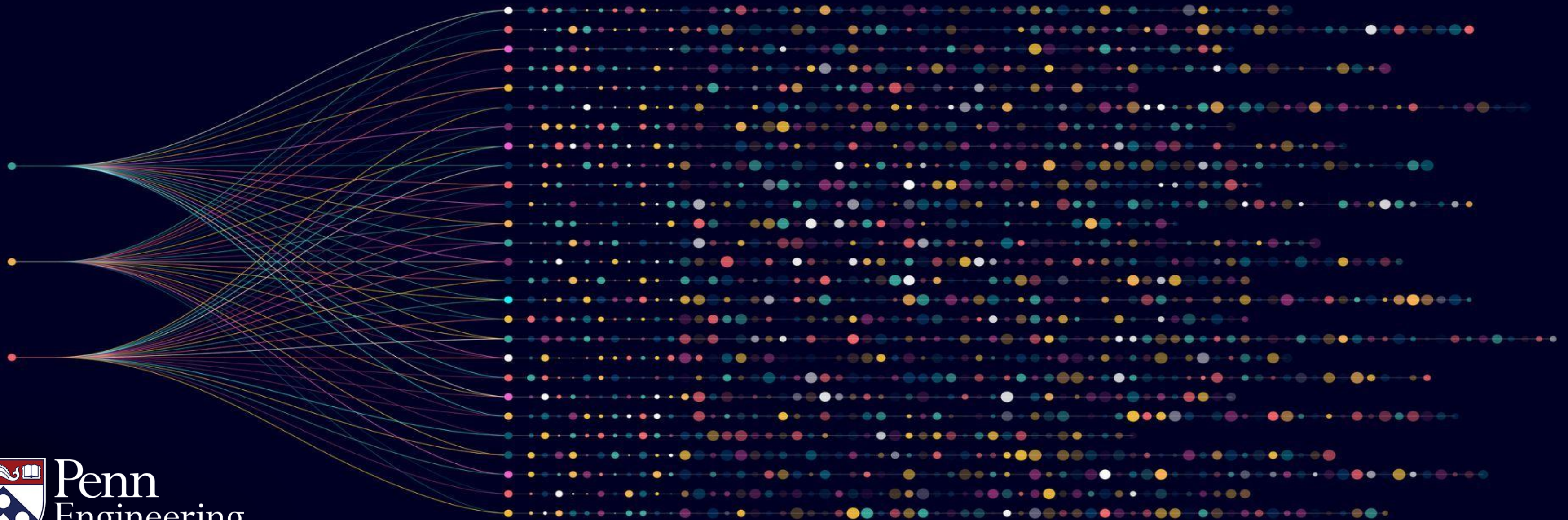


Algorithms for Big Data

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Random variables



Big data and probability

- Randomized algorithms are fundamental to modern data analysis
- Both the design and analysis of such algorithms crucially rely on probabilistic analysis
- Good grasp of the basics of probability theory is crucial for success!

What are random variables?

Suppose we perform some probabilistic experiment – example toss three fair coins

A random variable is a function whose value is completely determined by the outcome of the experiment – e.g., in the three coin tosses, the number of heads is a random variable

Example of random variable

Example – Toss three fair coins (fair coin means $\Pr[H] = \Pr[T] = 1/2$)

Space of outcomes is $\{HHH\}, \{HHT\}, \{HTH\}, \{HTT\}, \{THH\}, \{THT\}, \{TTH\}$ and $\{TTT\}$
(each with probability $1/8$)

Let **Count** denote the random variable which counts the number of heads in three coin tosses

$$\mathbf{Count} = \begin{cases} 3 \text{ w. p. } 1/8 \\ 2 \text{ w. p. } 3/8 \\ 1 \text{ w. p. } 3/8 \\ 0 \text{ w. p. } 1/8 \end{cases}$$

Random variables -- continued

We will mostly think of discrete valued random variables – i.e., our random variables take values from a discrete set (in fact, mostly a finite set)

For a random variable $\mathbf{X}$, we use $\Pr[\mathbf{X}=x]$ to denote the probability that the random variable $\mathbf{X}$ takes the value x

Note that $\sum_x \Pr[\mathbf{X}=x] = 1$. In other words, the sum of the probabilities over the various outcomes x is 1.

Expectation of a random variable

A fundamental quantity associated with a random variable is its expectation

We define the expectation of a random variable $\mathbf{X}$ as $E[\mathbf{X}] := \sum_x x \cdot \Pr [\mathbf{X} = x]$.

Note that the expectation of random variable $\mathbf{X}$ is its “average value” over the space of outcomes

Note: The expectation of a random variable $\mathbf{X}$ is also sometimes referred to as its mean

Expectation of a random variable – an example

Let us see an example of expectation calculation – in particular, consider **Count**, the random variable for counting the # of heads in three fair coin tosses.

$$\mathbf{Count} = \left\{ \begin{array}{l} 3 \text{ w. p. } 1/8 \\ 2 \text{ w. p. } 3/8 \\ 1 \text{ w. p. } 3/8 \\ 0 \text{ w. p. } 1/8 \end{array} \right\}$$

$$E[\mathbf{Count}] = 3 \cdot 1/8 + 2 \cdot 3/8 + 1 \cdot 3/8 + 0 \cdot 1/8 = 3/2 \text{ .}$$

Linearity of expectation

Arguably, the most fundamental fact about expectations of random variables is **linearity of expectation**.

Namely, for any two random variables $\mathbf{X}$ and $\mathbf{Y}$,

- I. $E[\mathbf{X} + \mathbf{Y}] = E[\mathbf{X}] + E[\mathbf{Y}]$ and
- II. For any $c \in \mathbb{R}$, $E[c \cdot \mathbf{X}] = c \cdot E[\mathbf{X}]$.

Linearity of expectation – a powerful tool

Linearity of expectation is a very powerful tool to compute expectations of random variables – even very complicated ones.

Example: Let $A = \{1, 2, 3, \dots, n\}$. Let $\mathbf{B}$ be a uniformly random permutation of A – i.e., a uniformly random rearrangement of A

So, if $n = 3$, then with probability $1/6$ each, $\mathbf{B} = \{1, 2, 3\}; \{1, 3, 2\}; \{2, 1, 3\}; \{2, 3, 1\}; \{3, 1, 2\}; \{3, 2, 1\}$

Fixed points of a permutation

For any permutation, the fixed points in a permutation are positions which are "unchanged"

Example: Permutation $\{1,2,3\}$ -- fixed points are 1, 2, and 3

Permutation $\{1,3,2\}$ -- fixed point is 1

Permutation $\{2,1,3\}$ -- fixed point is 3

Permutation $\{2,3,1\}$ -- no fixed points

Permutation $\{3,2,1\}$ -- fixed point is 2

Permutation $\{3,1,2\}$ -- no fixed points

Number of fixed points of a random permutation

Let $A = \{1, 2, 3, \dots, n\}$. Let $\mathbf{B}$ be a uniformly random permutation of A . Let $\text{FP}(\mathbf{B})$ denote the number of fixed points in $\mathbf{B}$.

Compute $E[\text{FP}(\mathbf{B})]$ – i.e., the expected number of fixed points of random permutation

Number of fixed points of a random permutation: attempt #1

Let $A = \{1, 2, 3, \dots, n\}$. Let $\mathbf{B}$ be a uniformly random permutation of A . Let $\text{FP}(\mathbf{B})$ denote the number of fixed points in $\mathbf{B}$.

Note that by the definition of expectation:

$$E[\text{FP}(\mathbf{B})] = \sum_{i=0}^n i \cdot \Pr[\text{FP}(\mathbf{B}) = i].$$

Complication: How to compute $\Pr[\text{FP}(\mathbf{B}) = i]$?

Attempt #2: exploiting linearity of expectation

Let $A = \{1, 2, 3, \dots, n\}$. Let $\mathbf{B}$ be a uniformly random permutation of A . Let $\text{FP}(\mathbf{B})$ denote the number of fixed points in $\mathbf{B}$.

Let $\text{FP}_j(\mathbf{B})$ denote the random variable which is 1 if j is a fixed point of $\mathbf{B}$ and 0 otherwise

Key observation: $\text{FP}(\mathbf{B}) = \sum_{j=1}^n \text{FP}_j(\mathbf{B})$

Linearity of expectation implies $E[\text{FP}(\mathbf{B})] = \sum_{j=1}^n E[\text{FP}_j(\mathbf{B})]$ — (1)

Computing $E[FP_j(\mathbf{B})]$

Let $FP_j(\mathbf{B})$ denote the random variable which is 1 if j is a fixed point of $\mathbf{B}$ and 0 otherwise

$$E[FP_j(\mathbf{B})] = 1 \cdot \Pr[FP_j(\mathbf{B}) = 1] + 0 \cdot \Pr[FP_j(\mathbf{B}) = 0] = \Pr[FP_j(\mathbf{B}) = 1]$$

Sidenote: A random variable $\mathbf{X}$ which takes only two values 0 and 1 is called an indicator (or Bernoulli) random variable

$$\text{For such an } \mathbf{X}, E[\mathbf{X}] = 1 \cdot \Pr[\mathbf{X} = 1] + 0 \cdot \Pr[\mathbf{X} = 0] = \Pr[\mathbf{X} = 1]$$

Computing $E[FP_j(\mathbf{B})]$

Let $FP_j(\mathbf{B})$ denote the random variable which is 1 if j is a fixed point of $\mathbf{B}$ and 0 otherwise

$$E[FP_j(\mathbf{B})] = \Pr[FP_j(\mathbf{B}) = 1]$$

For a random permutation $\mathbf{B}$, the index j is equally likely to get mapped to any of the positions in $\{1, 2, \dots, n\}$. Thus, $\Pr[FP_j(\mathbf{B}) = 1] = \frac{1}{n}$

This implies for any j , $E[FP_j(\mathbf{B})] = \frac{1}{n}$

Computing $E[\text{FP}(\mathbf{B})]$

Recall equation (1) -- $E[\text{FP}(\mathbf{B})] = \sum_{j=1}^n E[\text{FP}_j(\mathbf{B})]$

As each $E[\text{FP}_j(\mathbf{B})] = \frac{1}{n}$ for each index j , it follows that $E[\text{FP}(\mathbf{B})] = 1$

Thus, for a random permutation $\mathbf{B}$ of the set $A = \{1, 2, 3, \dots, n\}$, the expected number of fixed points is 1 (does not depend on n)

Fun way to reinterpret our calculation

Suppose n guests come to a party on a rainy afternoon. Each come with their umbrella

They put their umbrellas in an umbrella bucket. At the end of the party, the guests just randomly choose an umbrella from the bucket to take back with them 😊

Viewing the mapping between the umbrella and the guests as a random permutation of $A = \{1, 2, 3, \dots, n\}$, it means “on an average”, exactly 1 guest will get back their own umbrella.

Summary

- Random variables are functions of the underlying probabilistic experiment
- Also, we're looking at real-valued random variables (i.e., functions take value in $\mathbb{R}$) though generalizations are possible
- For a random variable $\mathbf{X}$, its expectation is the average value it takes over all possible outcomes of the underlying experiment

Summary

- Linearity of expectation:

$$E[X + Y] = E[X] + E[Y]$$

- Useful tool: Split a random variable X as
 $X = Z_1 + \cdots + Z_n$ where each Z_i is “simple”;

Analyze $E[Z_i]$ individually and obtain

$$E[X] = E[Z_1] + \cdots + E[Z_n]$$